TCP AND UDP COMPARISON

Task 13 | Cybersecurity Track
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1. Objective

To compare TCP and UDP communication and identify suitable use cases for each transport-layer protocol.

2. Introduction

TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are transport-layer protocols in the TCP/IP networking model. Both use port numbers to deliver data to applications, but differ in connection behavior, reliability, ordering, and overhead.

3. TCP vs UDP Comparison

Feature	TCP	UDP
Connection	Connection-oriented; establishes a connection before transferring data.	Connectionless; sends datagrams without establishing a connection.
Reliability	Acknowledgements, retransmission, and ordered delivery.	Does not guarantee delivery or ordering by itself.
Overhead	More protocol overhead; reliability mechanisms can add delay.	Lower protocol overhead and often lower latency.
Control	Includes flow control and congestion control.	No built-in TCP-style flow or congestion control.
Data format	Byte stream; data is delivered as a continuous stream.	Datagram-based; message boundaries are preserved.
Common uses	Many web connections, email, file transfer, SSH.	DNS queries, real-time voice/video, online gaming.

4. Appropriate Use Cases

TCP: Use when reliable, ordered delivery matters, such as file transfers, email, remote login, and many web connections.
UDP: Often useful when low latency is important, such as real-time voice/video, online games, and many DNS queries.

5. Advantages and Limitations

TCP advantages: reliable, ordered delivery and built-in flow/congestion control. **Limitation:** extra overhead and possible delay.
UDP advantages: low overhead and suitability for time-sensitive traffic. **Limitation:** delivery and order are not guaranteed by UDP itself.

6. Conclusion

TCP and UDP operate at the transport layer but serve different needs. TCP is appropriate when reliability and order matter. UDP is useful when low overhead and timeliness are priorities, provided the application can handle potential packet loss.

7. Interview Questions and Answers

Q1. What is TCP?
TCP is a connection-oriented transport protocol that provides reliable, ordered delivery of a byte stream.

Q2. What is UDP?
UDP is a connectionless transport protocol that sends datagrams with low overhead but does not guarantee delivery or order.

Q3. When is UDP preferred?
UDP is often preferred for real-time voice/video, online gaming, and other time-sensitive communication.

Q4. Which protocol is more reliable by itself?
TCP, because it provides acknowledgements and retransmissions.

Q5. Which protocol generally has lower overhead?
UDP.

References

1. RFC 9293 — Transmission Control Protocol (TCP).
2. RFC 768 — User Datagram Protocol (UDP).
3. Cloudflare Learning Center — TCP vs UDP overview.